A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 36

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	1 mark for the point below. **1** Cross-bred offspring are healthier / more productive because the problems caused by homozygous recessive genes in inbred varieties are reduced (heterosis) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
01.2	**1 mark each** 1 mark for each of the 2 points below. **1** Zebu tolerate heat but have a low milk yield – Ayrshire have a high milk yield but are not heat tolerant **1** Cross-breeding combined the desirable characteristics: heat tolerance with a high milk yield **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **tolerance** → the range of conditions in which a species can survive	2

Do not write outside the box

Question	Marking guidance	Marks
01.3	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Replacement of valuable animals that die with genetically identical individuals **1** Replacement of herds culled during a disease outbreak **1** Production of large numbers of individuals with desirable characteristics **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 321-322; AQA spec 3.5.1.2.	
02.1	**Any 2 of 6** 1 mark each for any 2 of the 6 points below, in any order, maximum 2. **1** Potato bean (N America): high protein content **1** Morama bean (southern Africa): protein content similar to soya, drought resistant **1** Yeheb tree (Somalia): edible nuts, drought resistant, grows on poor soils **1** Perennial maize (Mexico) / Kernza: no need to buy seed each year, less ploughing so less soil erosion **1 American bison:** graze grassland too poor for cattle **1 Common eland:** browse and graze so can be kept with cattle without severe competition **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'the soil loses its nutrients' as a description of erosion - erosion is the physical loss of the soil itself	2
02.2	**1 mark each** 1 mark for each of the 2 points below. **1** Indigenous species are better adapted to local conditions such as climate, pests / diseases and soil **1** So they have higher survival rates / productivity is not reduced, as it is when introduced species are kept in areas to which they are not adapted **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
02.3	1 mark for the point below. **1** They grow in soils with a high salt content, so their genes could be used in breeding programmes to increase the salt tolerance of crops grown on soils salinised by irrigation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **tolerance** → the range of conditions in which a species can survive	1

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 11; AQA spec 3.1.2.1.1.	
03.1	**1 mark each** 1 mark for each of the 3 points below. **1** Net held with its straight edge on the river bed facing the current (open to the flow) **1** The bed in front is disturbed by kicking / scraping a boot for a standard number of kicks and time (eg 30 s) **1** Same substrate type, water depth and flow rate at both sites – same net mesh – repeat samples – invertebrates identified with a key and counted **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **kick sampling** → disturbing the stream bed and catching the organisms carried downstream	3
03.2	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** Difficult to standardise the vigour / area kicked **1** Buried organisms or those fixed to stones are not dislodged **1** Some organisms swim and avoid the net **1** Flow rate varies and affects capture **1** Fragile organisms destroyed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **kick sampling** → disturbing the stream bed and catching the organisms carried downstream **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** A fixed area of bed is enclosed by a frame so the sampled area is standardised – side nets reduce escapes **1** But only works in fast enough flow and is hard to set on rough substrates / deep water **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **kick sampling** → disturbing the stream bed and catching the organisms carried downstream	2

Do not write outside the box

Question	Marking guidance	Marks
03.4	**1 mark each** 1 mark for each of the 2 points below. **1** Pollutant / oxygen levels fluctuate (eg with flow, time of day) so one reading may miss a pollution event **1** Sensitive taxa (stonefly, mayfly larvae) disappear and tolerant ones (midge larvae, hoglouse) remain, integrating conditions over weeks / months (biotic index) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biotic** → living factors / effects of other organisms	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 395-396, 415-416; AQA spec 3.7.2.1.	
04.1	**1 mark each** 1 mark for each of the 3 points below. **1** The new opportunity is overwhelmingly attractive (economic benefit) **1** Lack of knowledge / understanding of the impacts **1** The impacts occur elsewhere or much later, so are not evident to those who benefit **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
04.2	**1 mark each** 1 mark for each of the 2 points below. **1** Cultural belief that resources should be exploited to satisfy needs only **1** Problems of over-exploitation become obvious quickly as local supplies dwindle, so behaviour changes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** Resources are brought in from other areas **1** If one area is depleted, exploitation moves to another area (eg timber, fish, metals, energy) without the consumers noticing **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
04.4	**1 mark each** 1 mark for each of the 3 points below. **1** Easier access to energy has made many new activities possible **1** The scale of activities has greatly increased **1** Toxic wastes are produced that cannot be decontaminated by natural processes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-371; AQA spec 3.6.1.	
05.1	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** They are driven by renewable, low energy-density resources, especially solar energy, which will not be depleted: the hydrological cycle, the carbon and nitrogen cycles, atmospheric and thermohaline circulation **1** Most natural processes occur at low temperatures, because enzymes lower the activation energy of the reactions, so little energy is needed **1** Processes link into cycles in which the waste product of one is the raw material for the next, so raw materials are not depleted and wastes do not build up **1** A small number of abundant, simple elements are built into monomers and then into a wide range of polymers, so supplies of the raw materials are secure **1** Wastes are biodegradable and break down to non-toxic products, so they do not accumulate or poison the system **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs • **thermohaline** → the ocean conveyor driven by temperature and salinity • **biodegradable** → broken down by decomposers **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	3
05.2	**1 mark each** 1 mark for each of the 2 points below. **1** Minerals are dispersed after use - spread thinly through products all over the world, or lost to corrosion - so there is no concentrated deposit left to recover **1** Products are made of mixtures of materials that are difficult to separate, so recovery and re-use are expensive and often incomplete **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2

Do not write outside the box

Question	Marking guidance	Marks
05.3	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1 Design for extended use:** a product with a long life, which can be repaired, gives fewer opportunities for its materials to be lost **1 Design for end-of-life re-use:** materials that can be identified and separated easily are far more likely to be recycled **1** Separation of technical and biological materials, because metals and plastics need different reprocessing from biodegradable materials, which can be composted or digested **1** Energy from renewable sources at every stage, as in natural systems **1** Optimum rather than maximum production, so that the system supports the processes it relies on instead of maximising one output **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodegradable** → broken down by decomposers **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	3
05.4	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Sustainability depends on the total impact, which is the number of people multiplied by the impact of each person, so the same activity crosses the threshold as either rises **1** A natural process can absorb or replace what is taken only up to a rate – beyond that the resource is depleted or the waste accumulates faster than it can be broken down **1 Example:** slash and burn agriculture is sustainable where the population is small enough for each clearing to be left fallow long enough to recover, but not when land has to be re-cleared before the soil fertility has returned **1 Example:** discharging untreated sewage into a river is harmless where decomposers can break it down, but causes deoxygenation and eutrophication once the load exceeds that rate **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs • **threshold** → the concentration below which no harm is detected • **deoxygenation** → loss of dissolved oxygen from the water **Do not credit** • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	3

Do not write outside the box

Question	Marking guidance	Marks
05.5	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Urban populations are supplied by resources brought in from elsewhere, so the damage caused by extraction happens out of sight and often in another country **1** In a sedentary rural community the resource is local, so the problems of over-exploitation become obvious quickly as supplies dwindle, and a cultural belief in limiting exploitation to needs develops **1 Consequence:** when one area is exhausted, exploitation simply moves to another area and the consumer sees no change, so the total damage keeps growing - this has happened with timber, energy, metal minerals and fish **1 Consequence:** there is no pressure from consumers to change, so unsustainable practices continue until the resource fails on a large scale **1 Consequence:** the costs fall on the people and ecosystems where extraction happens, not on those who benefit, so the market price never reflects the true environmental cost **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-377; AQA spec 3.6.2-3.6.3.	
06.1	**1 mark each** 1 mark for each of the 3 points below. **1** Surveys monitor photosynthesis rates, biomass, soil water content and the spread of pests / diseases so inputs are applied only where needed **1** Yield mapping of different parts of a field controls fertiliser application in each area for future crops **1** This maximises yield while avoiding wastage / reducing leaching and nutrient pollution **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **leaching** → nutrients or ions washed down out of the soil / washed away by water	3
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** Machinery, pesticides, fertilisers, improved breeds and infrastructure (transport, refrigeration) are expensive **1** They are mainly available in affluent societies – poorer farmers cannot afford the inputs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	2

Do not write outside the box

Question	Marking guidance	Marks
06.3	1 mark for the point below. **1** Provides an income that allows producers to afford basic rights such as water, education, health care and food **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
06.4	**1 mark each** 1 mark for each of the 2 points below. **1** Nitrates are leached into water bodies used as sources of potable water **1** High nitrate levels cause blue baby syndrome (methaemoglobinaemia) and nitrates may be carcinogenic **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **leached** → nutrients or ions washed down out of the soil / washed away by water	2
06.5	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Anaerobic digestion by microbes in rice padi fields **1** Livestock intestines (ruminant digestion) **1** Anaerobic decay of manure / slurry **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 329-332; AQA spec 3.5.1.3-3.5.1.4.	
07.1	**1 mark each** 1 mark for each of the 4 points below. **1** k-selected: very slow growth, does not breed until 30, few eggs, so the population cannot replace losses **1** Fish reach catchable size (20) before they breed (30) so fishing can remove the entire future breeding population **1** Deep-water aggregations easily located and caught by trawling **1** Recovery takes decades so fisheries have had to be closed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	4

Do not write outside the box

Question	Marking guidance	Marks
07.2	**1 mark each** 1 mark for each of the 2 points below. **1** Herring breed at 2 years - the same age they become catchable - so there is always a breeding population and many young **1** Cod are catchable at 2 but do not mature until 4, so heavy fishing removes most fish before they breed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	2
07.3	**1 mark each** 1 mark for each of the 2 points below. **1** Larger numbers of smaller / younger fish are caught, or bigger boats, nets and sonar increase catch effort **1** Masks the decline of the breeding population until a sudden collapse **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
07.4	**1 mark each** 1 mark for each of the 2 points below. **1** Neither is a target species but both are caught in demersal trawls – skate now extinct in the Baltic and critically endangered elsewhere **1** Very late maturity (skate 11 years, Greenland shark 100-150 years, living over 400) and few young mean even low by-catch mortality exceeds replacement **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 341-342, 348-349; AQA spec 3.5.2.2.	
08.1	**1 mark each** 1 mark for each of the 3 points below. **1** Upwelling currents bring nutrients from the seabed to surface waters **1** 24-hour daylight in summer allows a huge phytoplankton bloom **1** Algae also grow under sea ice where krill shelter and feed – krill support most of the food web **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
08.2	**1 mark each** 1 mark for each of the 3 points below. **1** Krill is the main food of baleen whales, penguins, seals and fish so over-harvesting removes food from most of the web **1** Toothfish overfishing removes the food of sperm whales and elephant seals **1** Long-lining for toothfish catches and drowns albatrosses on the hooks (by-catch) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	3
08.3	**1 mark each** 1 mark for each of the 3 points below. **1** Ice on land stores 70% of the Earth's fresh water, keeping sea levels low **1** High albedo of ice reflects sunlight, limiting warming (reduction in ice area causes further warming) **1 Carbon sequestration:** carbon in algae eaten by krill sinks to the seabed in krill faeces **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **albedo** → reflectivity / the proportion of solar radiation reflected / how much sunlight is bounced back **Do not credit** • 'ice reflects the heat back' - albedo is the reflection of incoming short-wave solar radiation, before it is absorbed	3
08.4	1 mark for the point below. **1** Stronger winds spreading ice – more snowfall from warmer, moister air freezing onto slush – fresh meltwater from land forming a surface layer that freezes more easily (long-term trend still expected to be a decline) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 67-71; AQA spec 3.1.2.3.4.	

Do not write outside the box

Question	Marking guidance	Marks
09.1	**1 mark each** 1 mark for each of the 4 points below. **1** Less dead organic matter / leaf litter so less humus and slower soil formation – fewer nutrients returned **1** Loss of canopy and litter exposes soil to rain splash and wind **1** Decomposing roots no longer bind soil so erosion, gullying and landslides increase (especially on slopes) **1** Reduced organic matter and biota lower infiltration and water retention **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **humus** → dark, decomposed organic matter in soil • **infiltration** → water soaking into the soil surface **Do not credit** • 'trees give us the oxygen we breathe' as the main impact - credit carbon storage, soil protection, the water cycle and habitat loss	4
09.2	**1 mark each** 1 mark for each of the 4 points below. **1** Reduced interception and transpiration so less water returned to the atmosphere and less rainfall downwind **1 Increased runoff:** more flooding after rain, lower flows in dry periods – sedimentation of rivers **1** Increased albedo of cleared land (reflects more sunlight) **1** Reduced carbon sequestration and loss of the carbon reservoir (CO_2 released by burning / decay) contributing to climate change **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **transpiration** → water lost as vapour from the leaves • **albedo** → reflectivity / the proportion of solar radiation reflected / how much sunlight is bounced back **Do not credit** • 'trees give us the oxygen we breathe' as the main impact - credit carbon storage, soil protection, the water cycle and habitat loss • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	4
09.3	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Fragmentation of remaining forest into isolated patches with non-viable populations / edge effects **1** Loss of inter-species relationships (pollinators, seed dispersers) causing further declines **1** Loss of genetic resources / potential medicines before they are discovered **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'trees give us the oxygen we breathe' as the main impact - credit carbon storage, soil protection, the water cycle and habitat loss	2

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 368-369; AQA spec 3.5.3.3.	
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **In-situ:** species remain in natural conditions with the whole community and inter-species relationships – protected areas (SSSI, NNR, SAC, SPA, Ramsar, MCZ), land ownership, agri-environment schemes, habitat creation and design (area, shape, corridors, diversity), management of abiotic and biotic factors, plagioclimax maintenance, population control, reintroductions and rewilding (Yellowstone wolves, Scottish beavers) • **Limits of in-situ:** external threats (pollution, climate change), enforcement, cost, conflicts with land use, small fragmented populations • **Ex-situ:** captive breeding when in-situ is impossible or insufficient – selection criteria – difficulties keeping and breeding (habitat size, food, relationships, breeding triggers, group dynamics, gene pool / inbreeding, hybridisation) – techniques (stud books, cryopreservation, AI, embryo transfer, micropropagation, cloning) – seed banks – successes (Hawaiian goose, Iberian lynx, red kite, large blue) • **Limits of ex-situ:** expensive, few species, loss of wild behaviour, small gene pools, release requires suitable habitat and post-release support (hard vs soft release, monitoring) • **Judgement:** ex-situ is a temporary safeguard / support that only succeeds if in-situ habitat exists to return to – both needed, in-situ fundamental **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **abiotic** → non-living / physical factors • **biotic** → living factors / effects of other organisms	9
	• **plagioclimax** → a community held back from its climax by human activity **Do not credit** • captive breeding alone as conservation - credit needs reintroduction, habitat protection or genetic management as well	
	Model answer - Level 3 (8 marks)	

Do not write outside the box

Question	Marking guidance	Marks
	In-situ conservation keeps a species in its natural habitat with the whole community it depends on - food species, pollinators, predators and the abiotic conditions it is adapted to. It is delivered through protected areas such as SSSIs, NNRs, SACs, SPAs, Ramsar sites and Marine Conservation Zones, land ownership by conservation bodies, agri-environment schemes and habitat creation and management. Its weaknesses are that external threats such as pollution and climate change cross reserve boundaries, enforcement and cost, conflicts with other land uses, and the fact that many populations are already small and fragmented.	
	Ex-situ captive breeding is used when in-situ protection is impossible or insufficient. It is hard: animals need enough space, the right food, social groupings and breeding triggers, and small founder populations risk inbreeding and hybridisation, which studbooks, artificial insemination and embryo transfer help to manage. It is expensive, works for few species, and captive animals may lose wild behaviour; release succeeds only where suitable habitat exists and with soft release and post-release monitoring.	
	The two are not alternatives. Ex-situ methods are a temporary safeguard that can rescue a species from extinction, as with the Arabian oryx, but they only succeed if there is protected habitat to return to. In-situ conservation is therefore fundamental and ex-situ a support for it; the relative importance depends on how close to extinction the species is.	
	Weaker answer - Level 2 (5 marks)	
	In-situ conservation means protecting species in their natural habitat, for example in nature reserves and national parks. This is good because the species stays in the wild with the other species it needs. Ex-situ conservation means captive breeding in zoos and then releasing the animals back into the wild. This can save species that are nearly extinct but it is expensive and animals may not survive when released. Both methods are important and are often used together.	
	What makes the difference	
	• Level 2 defines both approaches correctly but Level 3 gives specific mechanisms (types of protected area, breeding difficulties, release methods) and named examples.	
	• Level 3 weighs the limitations of each approach explicitly; Level 2 gives one limitation for ex-situ only.	
	• 'Both are important' is not a judgement about *relative* importance; Level 3 explains why in-situ is fundamental and when ex-situ matters most.	
10.2	1 mark for the point below. **1** Restoring natural processes / communities to a habitat changed by humans, often by reintroducing missing (keystone) species and removing human management such as grazing **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 32-49; AQA spec 3.1.2.3.3-3.1.2.3.4.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Discuss how an understanding of energy flow and the manipulation of food species can be used to increase food production sustainably.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Energy flow:** autotrophs and heterotrophs – losses by reflection, transmission, respiration – 100 000 to 1 000 units – few food chains beyond four levels – plant food most productive – grazing animals where crops impossible – pigs on wastes • **Energetics:** intensive vs extensive, law of diminishing returns, energy subsidies (fertiliser, pesticides, irrigation, machinery, drying, processing, transport), energy ratios, food conversion ratios, dependence on fossil fuels • **Control of genetics:** asexual reproduction / cuttings / cloning, selective breeding and breeds, cross-breeding and hybrid vigour, AI and embryo transfer • **GM:** Roundup Ready soya, Bt crops, golden rice, omega-3 oilseed rape – advantages – risks (gene transfer, antibiotic markers, allergies, costs to LEDC farmers, loss of local diversity)	25
	• **Sustainability links:** energy dependence, trophic level of diet, feeding 12 billion people • Justified conclusion on the balance between productivity, efficiency and sustainability *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 321-328; AQA spec 3.5.1.2.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Discuss the extent to which agriculture can be made sustainable while feeding a growing human population.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Need:** population possibly 12 billion – humans raise carrying capacity – trophic level of diet • **Impacts to be reduced:** habitat loss, drainage, nutrient enrichment, introduced species, pesticide and nutrient pollution, nitrate toxicity, greenhouse gases, hydrological changes, soil erosion • **Social, economic and political influences:** consumer choices, technology availability, grants, guaranteed markets, surpluses, quotas, set-aside, agri-environment schemes • **Strategies:** cultural and biological pest control, integrated control, organic nutrient supply, legumes, low tillage, water conservation, CWR conservation, renewable energy, methane reduction, habitat retention • **Limits:** little new land, imports reducing local availability, cost and acceptance of new techniques, past land degradation • Justified conclusion *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 328-337; AQA spec 3.5.1.3-3.5.1.5.	25

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